



momently

API Service Contract



CYBERNAUTS

This document defines the API contracts for the Timesheets AI system. All endpoints return JSON responses and require appropriate authentication unless specified otherwise.

1. AUTHENTICATION API

1.1 Register New User

ENDPOINT: POST /api/auth/register

DESCRIPTION: Creates a new user account with the provided credentials. The email must end with @momentum.co.za domain. Password must contain at least 8 characters, one uppercase letter, one number, and one special character.

REQUEST BODY:

```
{
  "firstName": "John",
  "lastName": "Doe",
  "email": "john.doe@momentum.co.za",
  "password": "Secret@123"
}
```

RESPONSE (201 Created):

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "email": "john.doe@momentum.co.za",
  "firstName": "John",
  "lastName": "Doe",
  "createdAt": "2026-05-22T10:00:00Z",
  "message": "Registration successful! Please check your email for verification link."
}
```

RESPONSE (400 Bad Request - Validation Error):

```
{
  "status": 400,
  "error": "Bad Request",
  "message": "Email must end with @momentum.co.za"
}
```

1.2 Verify Email

ENDPOINT: `POST /api/auth/verify-email?token={verificationToken}`

DESCRIPTION: Verifies a user's email address using the token sent via email.
After successful verification, the user account becomes active.

REQUEST PARAMETERS:

- token: The verification token sent to user's email

RESPONSE (200 OK):

```
{
  "message": "Email verified successfully. You can now login."
}
```

RESPONSE (400 Bad Request - Invalid/Expired Token):

```
{
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid or expired verification token"
}
```

1.3 Login

ENDPOINT: POST /api/auth/login

DESCRIPTION: Authenticates a user and returns a JWT token for subsequent API calls. The token is valid for 1 hour.

REQUEST BODY:

```
{
  "email": "john.doe@momentum.co.za",
  "password": "Secret@123"
}
```

RESPONSE (200 OK):

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "expiresAt": "2026-05-22T11:00:00Z",
  "user": {
    "id": "123e4567-e89b-12d3-a456-426614174000",
    "email": "john.doe@momentum.co.za",
    "firstName": "John",
    "lastName": "Doe"
  }
}
```

RESPONSE (401 Unauthorized - Invalid Credentials):

```
{
  "status": 401,
  "error": "Unauthorized",
  "message": "Invalid email or password"
}
```

1.4 Forgot Password

ENDPOINT: POST /api/auth/forgot-password

DESCRIPTION: Sends a password reset link to the user's email address. Optional for Demo 1.

REQUEST BODY:

```
{  
  "email": "john.doe@momentum.co.za"  
}
```

RESPONSE (200 OK):

```
{  
  "message": "Password reset link sent to your email address"  
}
```

RESPONSE (404 Not Found - User Does Not Exist):

```
{  
  "status": 404,  
  "error": "Not Found",  
  "message": "No account found with this email address"  
}
```

1.5 Reset Password

ENDPOINT: POST /api/auth/reset-password

DESCRIPTION: Resets a user's password using a valid reset token.

REQUEST BODY:

```
{
  "token": "reset-token-from-email",
  "newPassword": "NewSecure@123"
}
```

RESPONSE (200 OK):

```
{
  "message": "Password reset successfully. You can now login with your new password."
}
```

RESPONSE (400 Bad Request - Invalid/Expired Token):

```
{
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid or expired reset token"
}
```

1.6 Logout

ENDPOINT: POST /api/auth/logout

DESCRIPTION: Invalidates the user's JWT token. Requires Authorization header.

REQUEST HEADERS:

- Authorization: Bearer {jwt-token}

RESPONSE (204 No Content):

(Empty response body)

2. TIME ENTRY API

2.1 Create Time Entry (Draft)

ENDPOINT: POST /api/time-entries

DESCRIPTION: Creates a new draft time entry. Only users with DEVELOPER role can create time entries. Requires workspace member ID in header.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

REQUEST BODY:

```
{
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174002",
  "startTime": "2026-05-22T09:00:00Z",
  "endTime": "2026-05-22T11:30:00Z",
  "durationMinutes": 150,
  "entryType": "manual",
  "description": "Worked on login page validation"
}
```

RESPONSE (201 Created):

```
{
  "id": "123e4567-e89b-12d3-a456-426614174010",
  "workspaceMemberId": "123e4567-e89b-12d3-a456-426614174100",
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174002",
  "startTime": "2026-05-22T09:00:00Z",
  "endTime": "2026-05-22T11:30:00Z",
  "durationMinutes": 150,
  "entryType": "manual",
  "description": "Worked on login page validation",
  "status": "DRAFT",
  "isLocked": false,
  "createdAt": "2026-05-22T09:00:00Z",
  "updatedAt": "2026-05-22T09:00:00Z",
  "rejectionReason": null
}
```

RESPONSE (400 Bad Request - Validation Error):

```
{
  "status": 400,
  "error": "Bad Request",
  "message": "Project ID is required"
}
```

RESPONSE (403 Forbidden - Not Assigned to Project):

```
{
  "status": 403,
  "error": "Forbidden",
  "message": "You are not assigned to this project"
}
```

2.2 Get My Time Entries

ENDPOINT: GET /api/time-entries/me

DESCRIPTION: Returns all time entries for the authenticated user.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

RESPONSE (200 OK):

```
[
  {
    "id": "123e4567-e89b-12d3-a456-426614174010",
    "workspaceMemberId": "123e4567-e89b-12d3-a456-426614174100",
    "projectId": "123e4567-e89b-12d3-a456-426614174001",
    "taskId": "123e4567-e89b-12d3-a456-426614174002",
    "startTime": "2026-05-22T09:00:00Z",
    "endTime": "2026-05-22T11:30:00Z",
    "durationMinutes": 150,
    "entryType": "manual",
    "description": "Worked on login page validation",
    "status": "DRAFT",
    "isLocked": false,
    "createdAt": "2026-05-22T09:00:00Z",
    "updatedAt": "2026-05-22T09:00:00Z",
    "rejectionReason": null
  }
]
```

2.3 Get Time Entries by Status

ENDPOINT: GET `/api/time-entries/me/status/{status}`

DESCRIPTION: Returns time entries for the authenticated user filtered by status.

Status can be: DRAFT, SUBMITTED, APPROVED, REJECTED.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

PATH PARAMETERS:

- status: DRAFT, SUBMITTED, APPROVED, or REJECTED

RESPONSE (200 OK):

```
[
  {
    "id": "123e4567-e89b-12d3-a456-426614174010",
    "workspaceMemberId": "123e4567-e89b-12d3-a456-426614174100",
    "projectId": "123e4567-e89b-12d3-a456-426614174001",
    "taskId": null,
    "startTime": "2026-05-22T09:00:00Z",
    "endTime": "2026-05-22T11:30:00Z",
    "durationMinutes": 150,
    "entryType": "manual",
    "description": "Worked on login page validation",
    "status": "SUBMITTED",
    "isLocked": false,
    "createdAt": "2026-05-22T09:00:00Z",
    "updatedAt": "2026-05-22T11:00:00Z",
    "rejectionReason": null
  }
]
```

RESPONSE (400 Bad Request - Invalid Status):

```
{
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid status value. Allowed: DRAFT, SUBMITTED, APPROVED, REJECTED"
}
```

2.4 Get Time Entry by ID

ENDPOINT: GET `/api/time-entries/{id}`

DESCRIPTION: Returns a specific time entry by its ID. Users can only access their own time entries. Managers can access entries of their team members.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

PATH PARAMETERS:

- id: The UUID of the time entry

RESPONSE (200 OK):

```
{
  "id": "123e4567-e89b-12d3-a456-426614174010",
  "workspaceMemberId": "123e4567-e89b-12d3-a456-426614174100",
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174002",
  "startTime": "2026-05-22T09:00:00Z",
  "endTime": "2026-05-22T11:30:00Z",
  "durationMinutes": 150,
  "entryType": "timer",
  "description": "Fixed authentication bug",
  "status": "APPROVED",
  "isLocked": true,
  "createdAt": "2026-05-22T09:00:00Z",
  "updatedAt": "2026-05-22T15:00:00Z",
  "rejectionReason": null
}
```

RESPONSE (404 Not Found):

```
{
  "status": 404,
  "error": "Not Found",
  "message": "Time entry not found with id:
123e4567-e89b-12d3-a456-426614174010"
}
```

2.5 Update Time Entry

ENDPOINT: PUT `/api/time-entries/{id}`

DESCRIPTION: Updates an existing draft or rejected time entry. Only the owner can update, and only if the entry is not locked or approved.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

PATH PARAMETERS:

- id: The UUID of the time entry

REQUEST BODY:

```
{
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174003",
  "startTime": "2026-05-22T10:00:00Z",
  "endTime": "2026-05-22T12:30:00Z",
  "durationMinutes": 150,
  "entryType": "manual",
  "description": "Updated: Worked on API integration"
}
```

RESPONSE (200 OK):

```
{
  "id": "123e4567-e89b-12d3-a456-426614174010",
  "workspaceMemberId": "123e4567-e89b-12d3-a456-426614174100",
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174003",
  "startTime": "2026-05-22T10:00:00Z",
  "endTime": "2026-05-22T12:30:00Z",
  "durationMinutes": 150,
  "entryType": "manual",
  "description": "Updated: Worked on API integration",
  "status": "DRAFT",
  "isLocked": false,
  "createdAt": "2026-05-22T09:00:00Z",
  "updatedAt": "2026-05-22T10:30:00Z",
  "rejectionReason": null
}
```

RESPONSE (403 Forbidden - Not Owner):

```
{
  "status": 403,
  "error": "Forbidden",
  "message": "You can only edit your own time entries"
}
```

RESPONSE (409 Conflict - Cannot Edit Locked/Approved):

```
{
  "status": 409,
  "error": "Conflict",
  "message": "Cannot edit time entry that is already APPROVED"
}
```

2.6 Delete Time Entry

ENDPOINT: DELETE /api/time-entries/{id}

DESCRIPTION: Deletes a draft time entry. Only the owner can delete, and only if the entry is still in DRAFT status.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

PATH PARAMETERS:

- id: The UUID of the time entry

RESPONSE (204 No Content):
(Empty response body)

RESPONSE (403 Forbidden - Not Owner):

```
{
  "status": 403,
  "error": "Forbidden",
  "message": "You can only delete your own time entries"
}
```

RESPONSE (409 Conflict - Cannot Delete Submitted/Approved):

```
{
  "status": 409,
  "error": "Conflict",
  "message": "Cannot delete time entry that is already SUBMITTED"
}
```

2.7 Submit Time Entry for Approval

ENDPOINT: **POST** `/api/time-entries/{id}/submit`

DESCRIPTION: Submits a draft time entry for manager approval. After submission, the entry cannot be edited by the developer.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

PATH PARAMETERS:

- id: The UUID of the time entry

RESPONSE (200 OK):

```
{
  "id": "123e4567-e89b-12d3-a456-426614174010",
  "workspaceMemberId": "123e4567-e89b-12d3-a456-426614174100",
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174002",
  "startTime": "2026-05-22T09:00:00Z",
  "endTime": "2026-05-22T11:30:00Z",
  "durationMinutes": 150,
  "entryType": "manual",
  "description": "Worked on login page validation",
  "status": "SUBMITTED",
  "isLocked": false,
  "createdAt": "2026-05-22T09:00:00Z",
  "updatedAt": "2026-05-22T11:00:00Z",
  "submittedAt": "2026-05-22T11:00:00Z",
  "rejectionReason": null
}
```

RESPONSE (409 Conflict - Already Submitted):

```
{
  "status": 409,
  "error": "Conflict",
  "message": "Time entry has already been submitted"
}
```

3. TIMER API

3.1 Start Timer

ENDPOINT: POST /api/timers/start

DESCRIPTION: Starts a new timer session for the user. Only one active timer per user is allowed. The timer automatically creates a time entry when stopped.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

REQUEST BODY:

```
{
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174002",
  "notes": "Starting work on authentication module"
}
```

RESPONSE (200 OK):

```
{
  "id": "123e4567-e89b-12d3-a456-426614174020",
  "projectId": "123e4567-e89b-12d3-a456-426614174001",
  "taskId": "123e4567-e89b-12d3-a456-426614174002",
  "startedAt": "2026-05-22T09:00:00Z",
  "active": true,
  "elapsedMinutes": 0
}
```

RESPONSE (409 Conflict - Timer Already Active):

```
{
  "status": 409,
  "error": "Conflict",
  "message": "You already have a running timer. Stop it before starting a new one.",
  "activeTimerId": "123e4567-e89b-12d3-a456-426614174019"
}
```

RESPONSE (400 Bad Request - Project ID Required):

```
{
  "status": 400,
  "error": "Bad Request",
  "message": "Project ID is required"
}
```

```
}
```

3.2 Stop Timer

ENDPOINT: POST /api/timers/stop

DESCRIPTION: Stops the active timer and automatically creates a DRAFT time entry with the calculated duration.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

RESPONSE (200 OK):

```
{
  "timerId": "123e4567-e89b-12d3-a456-426614174020",
  "stoppedAt": "2026-05-22T11:30:00Z",
  "durationMinutes": 150,
  "createdTimeEntry": {
    "id": "123e4567-e89b-12d3-a456-426614174021",
    "project": {
      "id": "123e4567-e89b-12d3-a456-426614174001",
      "name": "Mobile App Development"
    },
    "task": {
      "id": "123e4567-e89b-12d3-a456-426614174002",
      "title": "Implement authentication"
    },
    "date": "2026-05-22",
    "startTime": "09:00:00",
    "endTime": "11:30:00",
    "durationMinutes": 150,
    "status": "DRAFT"
  }
}
```

RESPONSE (404 Not Found - No Active Timer):

```
{
  "status": 404,
  "error": "Not Found",
  "message": "No active timer found"
}
```

3.3 Get Active Timer

ENDPOINT: GET `/api/timers/active`

DESCRIPTION: Returns the currently running timer for the user. Used to restore timer state after page refresh.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

RESPONSE (200 OK - Timer Running):

```
{
  "id": "123e4567-e89b-12d3-a456-426614174020",
  "project": {
    "id": "123e4567-e89b-12d3-a456-426614174001",
    "name": "Mobile App Development"
  },
  "task": {
    "id": "123e4567-e89b-12d3-a456-426614174002",
    "title": "Implement authentication"
  },
  "startedAt": "2026-05-22T09:00:00Z",
  "elapsedMinutes": 42,
  "active": true
}
```

RESPONSE (204 No Content - No Active Timer):
(Empty response body)

3.4 Discard Timer

ENDPOINT: DELETE /api/timers/discard

DESCRIPTION: Discards the active timer without creating a time entry. The timer session is deleted.

REQUEST HEADERS:

- X-Workspace-Member-Id: {workspace-member-uuid}

RESPONSE (204 No Content):

(Empty response body)

RESPONSE (404 Not Found - No Active Timer):

```
{
  "status": 404,
  "error": "Not Found",
  "message": "No active timer found"
}
```

4. INSIGHTS API

4.1 Get Insights Summary

ENDPOINT: GET `/api/insights/summary?from={date}&to={date}`

DESCRIPTION: Returns aggregated insights for the authenticated user within the specified date range. For managers, returns team-level insights.

REQUEST HEADERS:

- Authorization: Bearer {jwt-token}

QUERY PARAMETERS:

- from: Start date in YYYY-MM-DD format

- to: End date in YYYY-MM-DD format

RESPONSE (200 OK):

```
{
  "totalHoursLogged": 42.5,
  "averageHoursPerDay": 6.07,
  "totalDaysLogged": 7,
  "hoursPerProject": [
    {
      "projectId": "123e4567-e89b-12d3-a456-426614174001",
      "projectName": "Mobile App Development",
      "hours": 28.5,
      "entryCount": 12
    },
    {
      "projectId": "123e4567-e89b-12d3-a456-426614174005",
      "projectName": "API Gateway",
      "hours": 14.0,
      "entryCount": 6
    }
  ],
  "hoursPerTask": [
    {
      "taskId": "123e4567-e89b-12d3-a456-426614174002",
      "taskTitle": "Implement authentication",
      "hours": 18.5,
      "status": "IN_PROGRESS"
    },
    {
```

```

      "taskId": "123e4567-e89b-12d3-a456-426614174003",
      "taskTitle": "Write unit tests",
      "hours": 24.0,
      "status": "DONE"
    }
  ],
  "dailyTrend": [
    {
      "date": "2026-05-18",
      "hours": 6.5,
      "entryCount": 3
    },
    {
      "date": "2026-05-19",
      "hours": 5.0,
      "entryCount": 2
    },
    {
      "date": "2026-05-20",
      "hours": 8.0,
      "entryCount": 4
    }
  ]
}

```

RESPONSE (400 Bad Request - Invalid Date Range):

```

{
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid date range. 'from' must be before or equal to
'to'"
}

```

5. REPORTS API

5.1 Generate Productivity Report

ENDPOINT: GET `/api/reports/productivity?from={date}&to={date}`

DESCRIPTION: Generates a detailed productivity report for the authenticated user within the specified date range. For managers, generates team-level reports.

REQUEST HEADERS:

- Authorization: Bearer {jwt-token}

QUERY PARAMETERS:

- from: Start date in YYYY-MM-DD format
- to: End date in YYYY-MM-DD format

RESPONSE (200 OK):

```
{
  "generatedAt": "2026-05-22T15:30:00Z",
  "period": {
    "from": "2026-05-01",
    "to": "2026-05-31"
  },
  "summary": {
    "totalHoursLogged": 126.5,
    "totalEntriesLogged": 45,
    "tasksWorkedOn": 8,
    "projectsWorkedOn": 3
  },
  "byTask": [
    {
      "taskId": "123e4567-e89b-12d3-a456-426614174002",
      "taskTitle": "Implement authentication",
      "projectName": "Mobile App Development",
      "hoursLogged": 42.5,
      "entryCount": 15
    },
    {
      "taskId": "123e4567-e89b-12d3-a456-426614174003",
      "taskTitle": "Write unit tests",
      "projectName": "Mobile App Development",
      "hoursLogged": 38.0,
      "entryCount": 12
    }
  ]
}
```

```
    }  
  ],  
  "byWeek": [  
    {  
      "week": "2026-W18",  
      "hours": 32.5,  
      "entryCount": 12  
    },  
    {  
      "week": "2026-W19",  
      "hours": 45.0,  
      "entryCount": 18  
    },  
    {  
      "week": "2026-W20",  
      "hours": 49.0,  
      "entryCount": 15  
    }  
  ]  
}
```

6. COMMON DATA MODELS

6.1 TimeEntryStatus Enum

VALUES:

- DRAFT: Time entry is being edited, not yet submitted
- SUBMITTED: Time entry submitted for manager approval
- APPROVED: Time entry approved by manager
- REJECTED: Time entry rejected by manager, needs revision

6.2 EntryType Enum

VALUES:

- manual: Manually entered by user
- timer: Created by timer start/stop
- import: Imported from external source (Jira, calendar, etc.)
- api: Created via REST API

6.3 WorkspaceRole Enum

VALUES:

- ADMIN: Full workspace control, can manage users and settings
- MANAGER: Manages teams and projects, can approve timesheets
- DEVELOPER: Logs time, views own data only

6.4 User Entity

```
{
  "id": "uuid",
  "firstName": "string",
  "lastName": "string",
  "email": "string",
  "avatarUrl": "string",
  "timezone": "string",
  "jobTitle": "string",
  "seniorityLevel": "string",
  "employmentType": "string",
  "status": "string",
  "createdAt": "datetime",
  "updatedAt": "datetime"
}
```

6.5 Project Entity

```
{
  "id": "uuid",
  "workspaceId": "uuid",
  "name": "string",
  "description": "string",
  "status": "string",
  "visibility": "string",
  "budgetHours": "double",
  "budgetCost": "double",
  "hourlyRate": "double",
  "startDate": "datetime",
  "endDate": "datetime",
  "createdAt": "datetime",
  "updatedAt": "datetime"
}
```

6.6 Task Entity

```
{
  "id": "uuid",
  "projectId": "uuid",
  "title": "string",
  "description": "string",
  "statusId": "integer",
  "priority": "string",
  "estimatedHours": "double",
  "assignedWorkspaceMemberId": "uuid",
  "dueDate": "datetime",
  "completedAt": "datetime",
  "isDeleted": "boolean",
  "createdAt": "datetime",
  "updatedAt": "datetime"
}
```

6.7 TimeEntry Entity

```
{
  "id": "uuid",
  "workspaceMemberId": "uuid",
  "projectId": "uuid",
  "taskId": "uuid",
  "startTime": "datetime",
  "endTime": "datetime",
  "durationMinutes": "integer",
  "entryType": "string",
  "description": "string",
  "status": "string (enum)",
  "isLocked": "boolean",
  "lockedAt": "datetime",
  "editedAt": "datetime",
  "editedByWorkspaceMemberId": "uuid",
  "submittedAt": "datetime",
  "reviewedByWorkspaceMemberId": "uuid",
  "reviewedAt": "datetime",
  "rejectionReason": "string",
  "createdAt": "datetime",
  "updatedAt": "datetime"
}
```

7. HTTP STATUS CODES SUMMARY

CODE	MEANING	USAGE
200	OK	Request successful, returns data
201	Created	Resource successfully created
204	No Content	Request successful, no data to return
400	Bad Request	Validation failed, malformed request
401	Unauthorized	Authentication required or failed
403	Forbidden	Authenticated but not authorized
404	Not Found	Resource does not exist
409	Conflict	Business rule violation
500	Internal Server Error	Server-side error

8. AUTHENTICATION HEADER FORMAT

For endpoints requiring authentication (all except register, login, verify-email, forgot-password, reset-password), include the JWT token in the Authorization header:

Authorization: Bearer {jwt-token}

For endpoints requiring workspace context (time entries, timers), also include the workspace member ID:

X-Workspace-Member-Id: {workspace-member-uuid}